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Cenozoic remagnetization of the Paleozoic rocks in the Kitakami massif of northeast Japan, and its tectonic implications

Yo-ichiro Otofuji [a,*], Kazuhiro Takemoto [a], Haider Zaman [a], Yoshitomo Nishimitsu [a], Yutaka Wada [b]

[a] *Department of Earth and Planetary Sciences, Faculty of Science, Kobe University, Kobe 657-8501, Japan*
[b] *Department of Earth Sciences, Nara University of Education, Takabatake, Nara 630-8528, Japan*

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Abstract

Secondary remanent magnetization is identified in the Paleozoic igneous and sedimentary rocks of the Kitakami massif. The secondary nature is shown by a negative fold test for the Permo–Carboniferous sedimentary rocks. The northwesterly paleomagnetic declination with moderate inclination ($D = 321.2°$, $I = 56.5°$, $\alpha_{95} = 5.2°$, $N = 18$) of this secondary remanent magnetization is almost parallel to the primary magnetization reported for the Cenozoic welded tuffs of northeast Japan, indicating that the Paleozoic rocks were subjected to remagnetization at any period between 62 and 16 Ma. The secondary magnetization of the serpentinized ultramafic rocks is carried by magnetite, which grew in veins and mesh rims of serpentine, whereas the carrier of the magnetization in limestones is fine-grained pyrrhotite. Combining this with the previously reported remagnetization of the Kitakami granitic rocks, it is suggested that rocks in the Kitakami massif were subjected to crystallization remanent magnetization at low-temperature conditions. Since serpentinization requires fluid migration, one of the most likely events is the eastward lateral migration of water into the Kitakami massif. We postulate a Cenozoic suturing of the Kitakami massif with the Asian continent as a plausible tectonic event for this fluid migration.
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Keywords: paleomagnetism; rock magnetism; remagnetization; serpentine; pyrrhotite; CRM; northeast Japan

1. Introduction

Secondary remanent magnetization in rocks has often been disdained as merely an obstacle for geomagnetic study. It conceals completely or partly the primary remanent magnetization in the rocks which provides important information on geomagnetism at any geological time. Today, study of the secondary remanent magnetization is considered useful and of great geoscientific significance [1,2].

The Kitakami massif occupies the eastern half of northeast Japan (Fig. 1). Pre-Tertiary rocks are widely exposed in this massif [3,4]. Previous pa-

* Corresponding author. Tel.: +81-78-803-5730; Fax: +81-78-803-5757.
E-mail address: otofuji@kobe-u.ac.jp (Y.-i. Otofuji).

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leomagnetic studies from this area indicate that most of the Paleozoic and Mesozoic rocks were remagnetized and their remanent magnetizations directed northwestward [5–7]. Because Paleozoic and Mesozoic rocks in the Kitakami massif preserve a well-defined secondary magnetization, rock and paleomagnetic magnetic study of these rocks can provide an important clue to determine the age of remagnetization and infer the tectonic event which brought about remagnetization in the Kitakami massif. We therefore present a new paleomagnetic result obtained from the Paleozoic and Mesozoic rocks of the Kitakami massif. The aim of this study is to find a reason for fluid migration-related remagnetization in the tectonic framework of northeast Japan.

2. Geological setting and sampling

Pre-Tertiary strata in the Kitakami massif have been divided into northern and southern terranes by the Hayachine tectonic belt (HTB) of mafic to ultramafic rocks [3,4,7] (Fig. 1). The northern Kitakami terrane (NKT) consists of the Jurassic accretionary complex and the southern Kitakami terrane (SKT) represents a pre-Cretaceous shallow marine sequence. These two terranes collided with each other at about 140 Ma and amalgamated to form the Kitakami massif [9].

The Hayachine–Miyamori district consists of the HTB and the northeastern margin of the SKT [10]. The HTB is characterized by an abundance of mafic to ultramafic rocks referred to the Hayachine and Miyamori complexes. K–Ar dating of gabbros and hornblendite places the age of these rocks in the range from 421 to 484 Ma [11,12], suggesting that these complexes were formed as a basal ultramafic portion of an ophiolite in Ordovician time. The ultamafics are strongly serpentinized and most of the rocks contain more than 60% serpentine [13]. In addition to the ultramafics, the Hisadenaigawa Metamorphic Rocks are thinly distributed within the Hayachine complex. The uppermost members of this complex are the Koguro and Yakushigawa formations. The Koguro formation comprises dolerites and basalts while the Yakushigawa formation includes terrigenous sandstones and shales. The Hayachine complex is overlain by the Silurian to Permian formations of the SKT, which consist of sandstone, shale, conglomerate, subordinate amounts of limestones and pyroclastic rocks [14]. The Miyamori complex thrusts southwestwards over these Paleozoic formations.

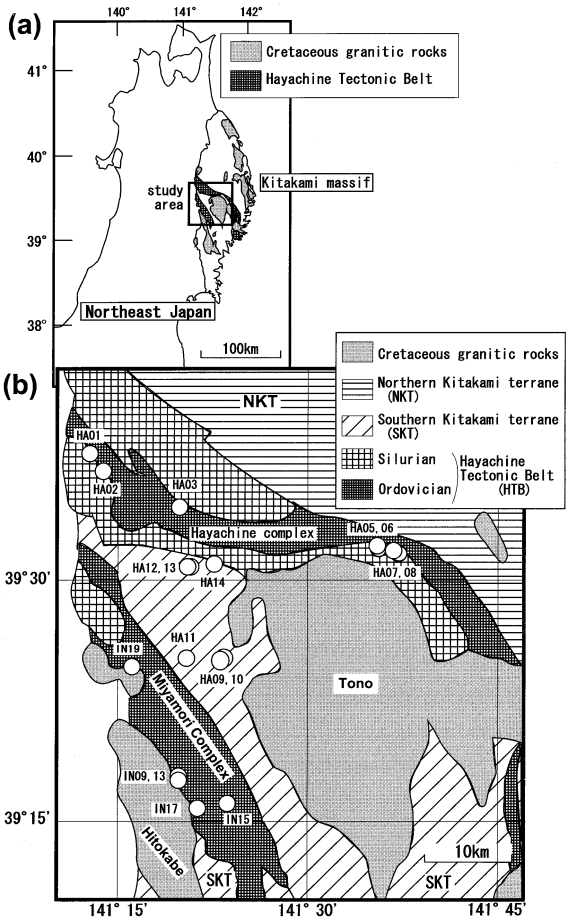


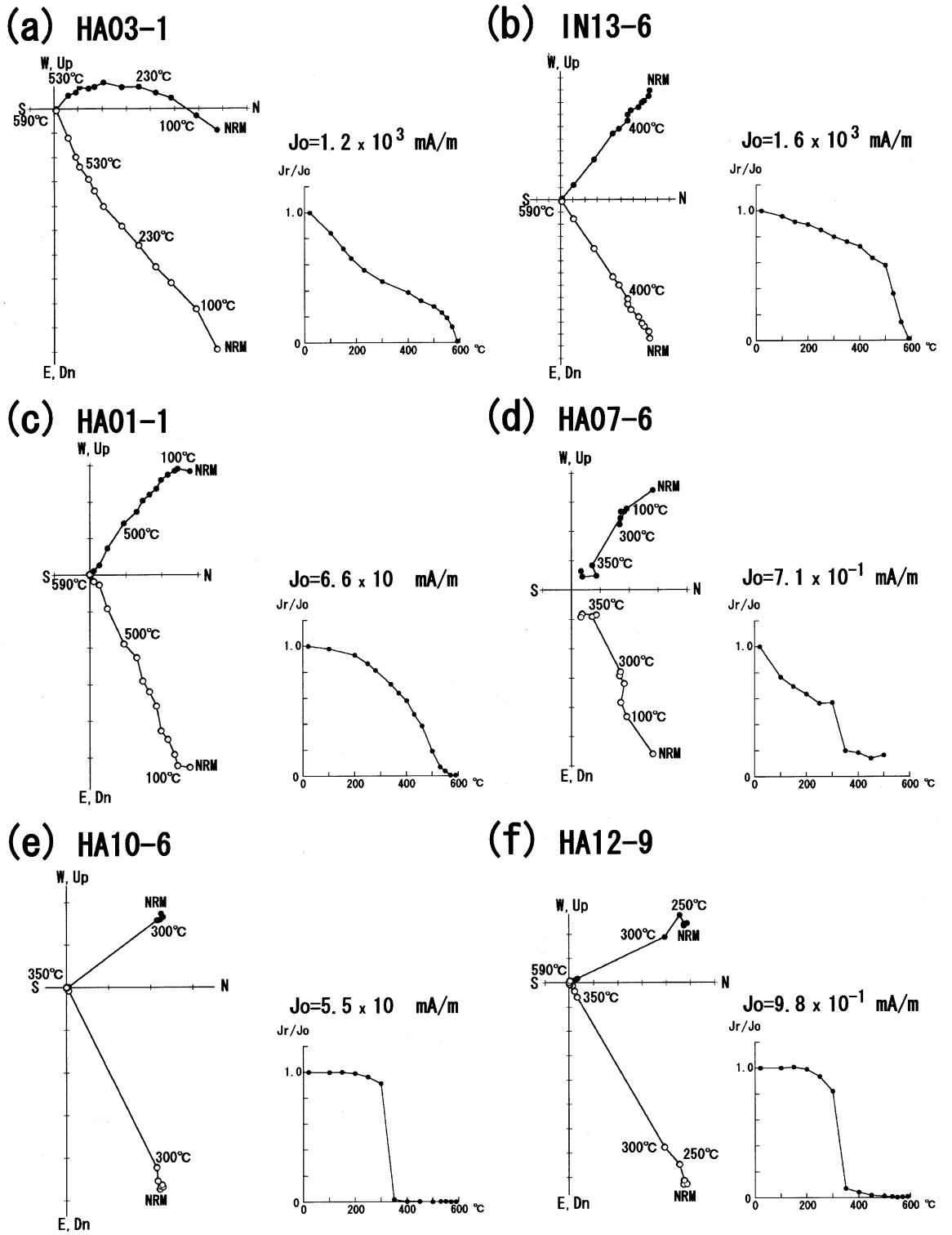
Fig. 1. (a) Location map of the study area within the framework of northeast Japan. Enlargement of the bracketed area is shown in panel b. (b) Geological and tectonic map of the study area. This map is modified from the geological maps published by the Editorial Committee of Tohoku [4] and Ehiro [8]. The northern Kitakami and southern Kitakami terranes (NKT and SKT, respectively) are divided by the Hayachine tectonic belt (HTB). Locations of the sampling sites are marked by open circles.

Late Mesozoic magmatism began in the early Cretaceous (109–135 Ma) throughout the SKT

Table 1
Paleomagnetic results of the Paleozoic rocks from the south Kitakami terrane (SKT) and Hayachine tectonic belt (HTB)

Age	Block	Formation	Site	Locality		*n*/*N*	THD	In situ		Tilt-corrected		*k*	α_{95}
				Lat. (N)	Long. (E)			*D* (°)	*I* (°)	*D* (°)	*I* (°)		(°)
Permian	SKT	Sakamotozawa Fm.	HA09 limestone	39°25′21″	141°22′57″	9/10	Po	321.3	73.2	22.1	63.7	11.0	16.2
			HA10 limestone	39°25′27″	141°23′05″	10/10	Po	316.5	62.9	279.4	−8.8	334.2	2.6
			HA11 limestone	39°25′24″	141°20′36″	10/10	Po	354.6	72.6	99.4	11.0	6.5	20.6
Carboniferous	SKT	Hikoroich Fm.	HA12 limestone	39°31′12″	141°20′28″	10/10	Po	310.3	51.5	7.3	10.2	5.5	22.7
			HA13 meta-sandstone	39°31′40″	141°21′57″	7/10	β	340.6	57.8	1.4	28.1	5.3	29.1
			HA14 meta-coarse s.s.	39°31′43″	141°22′10″	10/10	Po	309.4	54.7	320.5	42.5	67.9	5.9
	SKT mean			39.5°	141.4°	6		322.7	63.0			44.6	10.1
						6				356.7	38.5	2.1	62.3
Silurian	HTB	Yakushigawa Fm.	HA05 meta-mudstone	39°32′13″	141°35′37″	9/9	Po	340.3	79.5	–	–	3.7	31.0
			HA06 meta-mudstone	39°32′03″	141°35′37″	9/9	β	321.3	39.8	–	–	5.2	24.9
			HA07 meta-mudstone	39°31′56″	141°36′40″	7/10	Po	315.9	45.3	–	–	42.6	9.4
			HA08 meta-basic rock	39°31′49″	141°37′01″	9/10	β	330.5	52.5	–	–	11.7	15.7
	HTB	Koguro Fm.	HA01 meta-mafic rock	39°38′01″	141°12′24″	9/10	β	312.4	49.7	–	–	312.0	2.9
			HA02 meta-mafic rock	39°37′00″	141°14′32″	9/10	β	316.7	49.1	–	–	3.3	33.3
		mean		39.5°	141.4°	6		320.3	52.7			30.7	12.3
Ordovician	HTB	Miyamori complex	IN09 pyroxinite	39°17′31″	141°19′30″	29/29	β	330.4	52.8	–	–	76.8	3.1
			IN13 serpentine	39°17′42″	141°19′30″	10/11	β	311.2	44.1	–	–	139.9	4.1
			IN15 serpentine	39°16′08″	141°23′29″	12/12	β	330.1	58.3	–	–	147.4	3.6
			IN17 serpentine	39°15′52″	141°21′10″	12/12	β	327.4	50.4	–	–	21.0	9.7
			IN19 serpentine	39°24′00″	141°16′00″	10/12	β	309.2	57.8	–	–	52.7	6.0
	HTB	Hayachine complex	HA03 serpentine	39°34′42″	141°20′25″	9/9	β	318.0	56.4	–	–	49.3	7.4
		mean		39.5°	141.4°	6		320.8	53.7	–	–	107.5	6.5
	HTB mean			39.5°	141.4°	12		320.6	53.2	–	–	52.5	6.0
Overall mean				39.5°	141.4°	18		321.2	56.5	–	–	44.7	5.2

The magnetization directions carried by pyrrhotite and magnetite are listed as Po and β, respectively. THD = thermal demagnetization; *N* and *n* = numbers of samples collected and samples used for statistics, respectively; *D* and *I* = declination and inclination, respectively; *k* = the Fisherian precision parameter; α_{95} = the radius of the cone of 95% confidence.


(a) HA03-1
W, Up
530℃
230℃
S
N
590℃
100℃
NRM
530℃
230℃
100℃
NRM
E, Dn
Jo=1.2 x 10^3 mA/m
Jr/Jo
1.0
0
0
200
400
600 ℃
(b) IN13-6
W, Up
NRM
400℃
S
N
590℃
400℃
NRM
E, Dn
Jo=1.6 x 10^3 mA/m
Jr/Jo
1.0
0
0
200
400
600 ℃
(c) HA01-1
W, Up
100℃
NRM
500℃
S
N
590℃
500℃
NRM
100℃
E, Dn
Jo=6.6 x 10 mA/m
Jr/Jo
1.0
0
0
200
400
600 ℃
(d) HA07-6
W, Up
NRM
100℃
300℃
350℃
S
N
350℃
300℃
100℃
NRM
E, Dn
Jo=7.1 x 10^-1 mA/m
Jr/Jo
1.0
0
0
200
400
600 ℃
(e) HA10-6
W, Up
NRM
300℃
350℃
S
N
300℃
NRM
E, Dn
Jo=5.5 x 10 mA/m
Jr/Jo
1.0
0
0
200
400
600 ℃
(f) HA12-9
W, Up
250℃
300℃
NRM
590℃
S
N
350℃
300℃
250℃
NRM
E, Dn
Jo=9.8 x 10^-1 mA/m
Jr/Jo
1.0
0
0
200
400
600 ℃

and NKT [15]. Early Cretaceous Tono and Hitokabe batholiths intruded the western margin of the Hayachine and Miyamori complexes, respectively. The western part of the Miyamori complex experienced the effect of contact metamorphism.

Ordovician to Permian rocks were collected at 18 sites as oriented hand samples using a magnetic compass. Thirteen sites are from the Ordovician Hayachine and Miyamori complexes, and the remaining sites are from the Carboniferous Hikoroichi and the Permian Sakamotozawa formations of the SKT. The lithology of the collected samples consists of serpentinized ultramafic rocks, mafic rocks, sand–mud stones and limestones. Sampling localities, stratigraphic positions and rock types are shown in Fig. 1 and numerical details in Table 1. Each paleomagnetic site is typically comprised of 10 independently oriented samples, distributed over a distance of up to 15 m. Bedding attitude was only observed at sites collected from the SKT.

3. Laboratory procedure

Oriented hand samples were cored in the paleomagnetic laboratory of Kobe University. Individual specimens, 25 mm in diameter and 22 mm long, were prepared from these samples. Natural remanent magnetizations (NRMs) were measured with either a spinner magnetometer (Natsuhara SMM-85) or a cryogenic magnetometer (2G-Enterprise). Stepwise thermal demagnetization was carried out up to 690°C with a laboratory-built furnace or a Natsuhara TDS-1 thermal demagnetizer; the residual magnetic field in these furnaces is less than 5 nT. Results for each specimen were plotted on orthogonal vector diagrams [16] to assess component structure as well as on equal-area projections to evaluate directional stability. Principal-component analysis [17] was used to estimate the component directions.

4. Natural remanent magnetization

Thermal demagnetization procedure revealed that sedimentary rocks are generally characterized by a low unblocking temperature of about 350°C, whereas the mafic and serpentinized ultramafic rocks show unblocking temperatures between 550 and 590°C.

4.1. Ordovician ultramafic rocks

Ultramafic rocks of five sites (HA03, IN13, IN15, IN17, IN19) are completely or partially serpentinized, while rocks of the remaining site (IN09) consist of pyroxinites. All samples gave high initial NRM intensities, ranging from 1.2 to 15 A/m, and generally revealed two components of magnetization (Fig. 2a,b). The low-temperature component, however, is sometimes difficult to identify by principal-component analysis. The high-temperature component appears after the demagnetization level of 450–500°C and reveals straightforward demagnetization behavior with unblocking temperatures between 560 and 590°C.

The directions of the high-temperature component show tight clustering within all six sites, where the precision parameter ranges from 21.0 to 147.4 (Table 1). Site mean paleomagnetic directions of these sites reveal only normal polarity and give an approximately westerly direction with moderate inclination: $D = 309.2°$–$330.4°$ and $I = 44.1°$–$58.3°$. The in situ mean direction of six sites is $D = 320.8°$, $I = 53.6°$, $\alpha_{95} = 6.5°$ (Fig. 3). This direction is sub-parallel to the preliminary results of alternating field (AF) demagnetization [18,19].

4.2. Silurian igneous and sedimentary rocks

Initial NRM intensities of mafic rocks (HA01, HA02, HA08) varied between 1.0 and 4.7 A/m.

←

Fig. 2. Orthogonal projections of the vector end points during thermal demagnetization experiments. Ordovician serpentines from the HTB (panels a and b; HA03, IN13), Silurian rocks from the HTB (panels c and d; HA01, HA07), and Permo–Carboniferous limestones from the SKT (panels e and f; HA10, HA12). Solid (open) symbols are for horizontal (vertical) projection. All diagrams are in the geographic coordinates.

Each site exhibited fairly uniform demagnetization behavior and showed two components of magnetization in visual inspection (Fig. 2c,d). The low-temperature component was removed at 300°C, and then a high-temperature component was unblocked at temperatures between 550 and 590°C.

Meta-sedimentary rocks (HA05, HA06, HA07) gave relatively lower initial NRM intensities (0.1–0.7 A/m). Samples from sites HA05 and HA07 possess only a low-temperature component, with unblocking temperature ranging from 350 to 400°C. Samples of site HA06 possess two components, where the low- and high-temperature components were unblocked at 250 and 550–590°C, respectively.

Site mean directions of the low-temperature component for HA05 and HA07 as well as the high-temperature component in the remaining sites are characterized by a westerly declination and moderate inclination. The in situ mean direction of both these components for six sites is $D = 320.3°$, $I = 52.7°$, $\alpha_{95} = 12.3°$ (Fig. 3).

4.3. Carboniferous and Permian sedimentary rocks

Samples from limestones and sandstones gave initial NRM intensities in a range between 0.05 and 100 mA/m, while their unblocking temperatures are consistently approximately 350°C. In the case of HA09, HA10 and HA12, more than 90% of the NRM intensity was removed between 300 and 350°C (Fig. 2e,f). Samples of HA11 and HA14 also revealed a sharp decrease in their NRM intensity at 350°C, but lower intensities of magnetization remained up to an unblocking temperature of 590°C.

Two components of magnetization were observed in the samples of HA13, where initial NRM intensities were found in a range between 0.8 and 150 mA/m. The low-temperature component was demagnetized between 300 and 400°C, while the high-temperature component was present up to demagnetization temperatures of 550°C.

Five site mean directions with an unblocking temperature of 350°C (HA09, HA10, HA11, HA12, HA14) and one from the high-temperature component (HA13) show westerly declinations and moderate inclinations in geographic coordinates. The in situ mean direction from six sites is $D = 322.7°$, $I = 63.0°$, $\alpha_{95} = 10.1°$. After tilt cor-

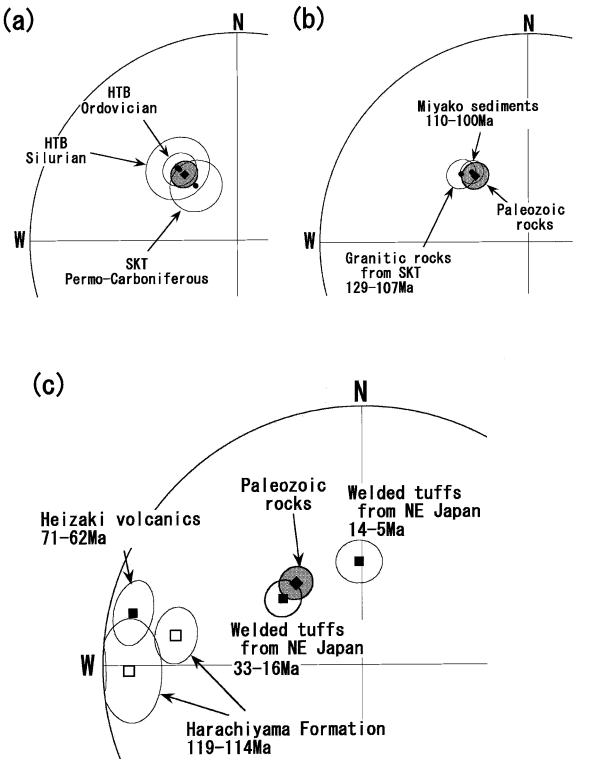


Fig. 3. Equal-area projections of paleomagnetic directions with a circle of 95% confidence for the Paleozoic to the Cenozoic rocks of the Kitakami massif and northeast Japan. (a) Paleomagnetic directions of the Hayachine tectonic block (HTB) and southern Kitakami terrane (SKT). Mean paleomagnetic direction of the Paleozoic rocks from the HTB and SKT is shown as a shaded circle (Table 1). (b) Mean paleomagnetic direction of the Paleozoic rocks from the HTB and SKT (shaded) is compared with in situ paleomagnetic directions of 110–100 Ma Miyako Cretaceous sediments ($D = 321.00°$, $I = 54.5°$, $\alpha_{95} = 0.5°$) and 129–107 Ma granite of the Kitakami massif ($D = 315.3°$, $I = 52.2°$, $\alpha_{95} = 5.7°$) [6,28]. (c) Mean paleomagnetic direction of the Paleozoic rocks from HTB and SKT (shaded) is compared with tilt-corrected paleomagnetic directions of 119–114 Ma Harachiyama volcanics ($D = 273.3°$, $I = -28.2°$, $\alpha_{95} = 7.7$; $D = 268.8°$, $I = -11.1°$, $\alpha_{95} = 12.7°$) and 71–62 Ma Heizaki volcanics ($D = 283.1°$, $I = 10.3°$, $\alpha_{95} = 8.0°$) [24], 33–16 Ma welded tuffs from northeast Japan ($D = 310.3°$, $I = 57.2°$, $\alpha_{95} = 5.7°$) [23] and 14–5 Ma welded tuffs from northeast Japan ($D = 358.2°$, $I = 57.1°$, $\alpha_{95} = 7.0°$) [23]. Solid (open) symbols refer to the lower (upper) hemisphere.

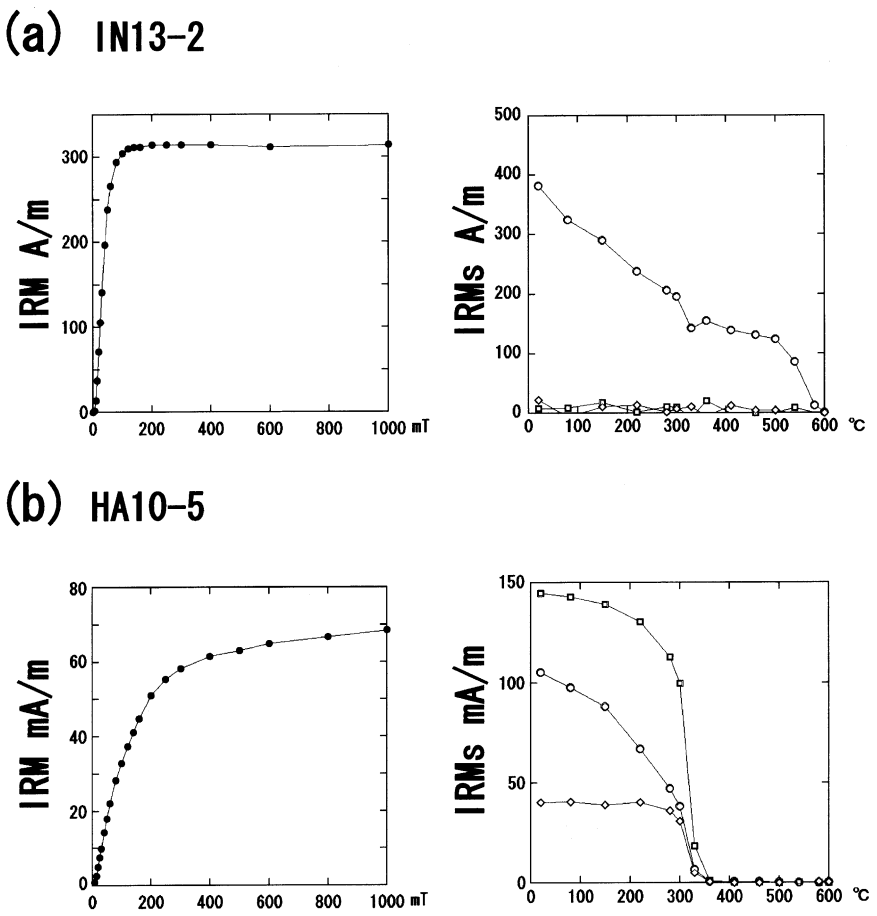


Fig. 4. IRM acquisition curve and thermal demagnetization of three-component IRMs for (a) serpentine (HA03) and (b) limestone (HA10). IRMs of different DC fields (◇ 2.7 T, □ 0.4 T and ○ 0.12 T) were imparted to each of three perpendicular axes of a specimen and thermally demagnetized. (a) The IRM acquisition curve for serpentine reaches a saturation point at 0.3 T DC field and the soft-coercivity fraction is unblocked at about 580°C. (b) The IRM acquisition curve for limestone indicates that no saturation is achieved up to 1.0 T and the unblocking temperature for all three coercivity fractions is around 330°C.

rection, clustered site mean directions became more dispersed. The ratio of the precision parameter before and after tilt correction ($k_2/k_1 = 2.1/44.6$) gives a negative fold test at the 99% confidence level [20]. This type of behavior indicates that the characteristic remanent magnetizations (ChRMs) of the Permo–Carboniferous rocks are secondary magnetizations acquired after tilting.

5. Rock magnetic study and interpretation

Rock magnetic experiments were conducted on samples from all sites to elucidate the carrier of remanent magnetization (Figs. 4 and 5). We performed AF demagnetization, progressive acquisition of isothermal remanent magnetization (IRM) and thermal demagnetization of three-component (3D) IRMs [21]. AF demagnetization was carried out with a three-axis tumbler system in a threefold cylindrical μ-metal shield.

Different rock magnetic parameters strongly suggest that the carrier of magnetization in the Ordovician serpentinized ultramafic rocks (HA03, IN13, IN15, IN17, IN19) is magnetite. The maximum unblocking temperatures of

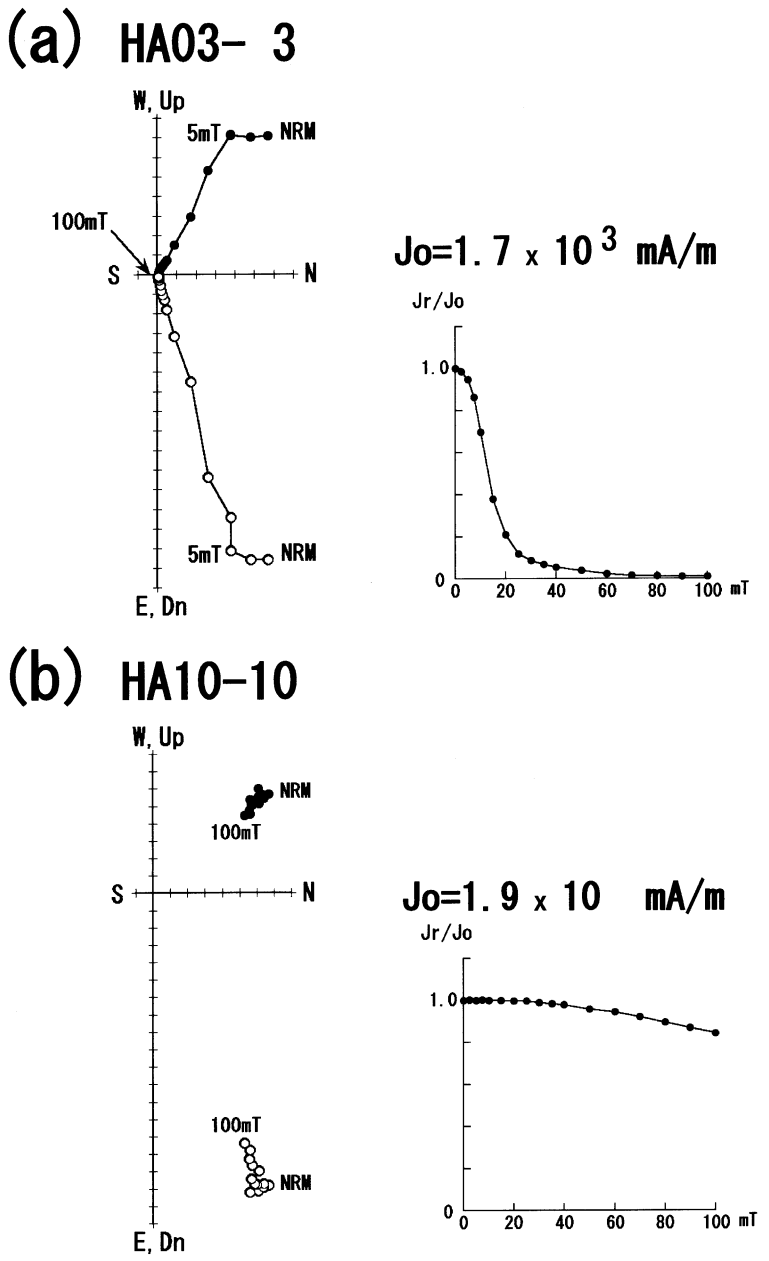


Fig. 5. Orthogonal projections of the vector end points during AF demagnetization experiments. (a) Ordovician serpentine from site HA03 and (b) Permian limestone from site HA10. Solid (open) symbols are for horizontal (vertical) projection. All diagrams are in the geographic coordinates.

580°C obtained from the NRM and 3D IRMs indicate magnetite as the remanence carrier (Figs. 2a,b and 4a). The presence of magnetite as predominant magnetic carrier is also confirmed by the low-coercivity response to AF demagnetization and IRM acquisition (Figs. 4a and 5a). A median destructive field of less than 20 mT is observed for these rocks. The process of IRM ac-

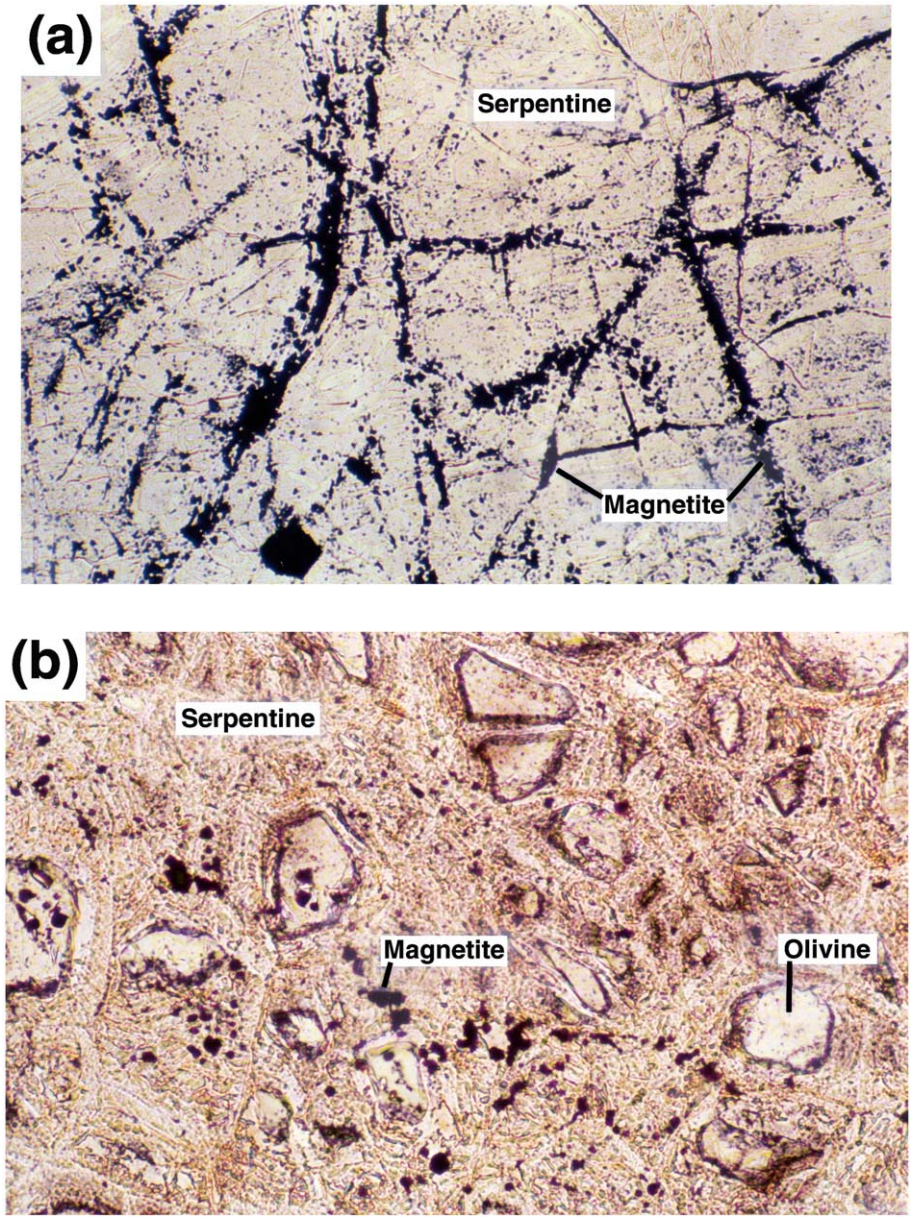


Fig. 6. Transmitted light photomicrographs of completely and partially serpentinized peridotites. (a) Completely serpentinized peridotite from site HA03 shows a typical serpentine mesh texture. (b) Partially serpentinized peridotite from site IN13 indicates olivine relics in a serpentine mesh. Magnetite appears in some places of the serpentine mesh rims. Lengths of the diagonal are 1.8 mm for (a) and 0.75 mm for (b).

quisition shows a rapid rise below 200 mT and complete saturation at about 300 mT. In addition to predominant magnetite, the presence of pyrrhotite can be suggested by a slight drop in thermal demagnetization of the soft IRM component at 325°C (Fig. 4a). Thermomagnetic analysis of serpentinized Miyamori ultramafic rocks [19] showed that, apart from predominant magnetite, trace amounts of pyrrhotite are also present.

Based on the unblocking temperature of 330°C

in thermal demagnetization plots (Fig. 2e,f), pyrrhotite is considered the main magnetic carrier of the Permo–Carboniferous limestones (HA09, HA10, HA11, HA12) and coarse-grained sandstones (HA14). The presence of pyrrhotite is also very clearly indicated by the thermal demagnetization of 3D IRMs, where the medium-coercivity fraction drops sharply at 330°C (Fig. 4b). These samples show high stability in response to AF demagnetization and IRM acquisition (Figs. 4b and 5b). This type of behavior is probably due to the presence of pyrrhotite, which possesses a higher coercive force than magnetite [22].

NRMs of the Silurian meta-sedimentary and felsic rocks are carried by pyrrhotite and/or magnetite, as shown by the thermal demagnetization diagrams (Fig. 2c,d).

6. Microscopic observation

We have studied nine thin sections of serpentinized ultramafic rocks and limestones under transmitted and reflected light (Fig. 6). Minerals in limestone were analyzed with a JEOL JSM-6330F scanning electron microscope equipped with an Oxford energy-dispersive analytical system (EDS; Link ISIS series 3.34) at Kobe University. Accelerating voltage and beam current were kept at 15.0 kV and 0.4 nA, respectively.

The extensively serpentinized ultramafic rocks (HA03, IN17) display typical serpentine mesh textures (Fig. 6a). Serpentine mesh size varies between 0.2 and 1 mm. Mesh centers apparently are made of amorphous serpentine and rims are mostly occupied by magnetite. In these samples, orthopyroxene is replaced by bastite. Pyrite and pyrrhotite are also present. In samples that are serpentinized to a moderate degree (IN13, IN15, IN19), serpentine isolates rounded olivine relics and magnetite is occasionally distributed in mesh rims and veins (Fig. 6b). The presence of brucite in serpentines is confirmed in the sample of IN15. The absence of antigorites in serpentine indicates that the studied rocks experienced no contact metamorphism.

In the limestone samples (HA09, HA10, HA11), round framboids of 5–20 μm (that consist of aggregates of small crystals) are clearly visible. These framboids were possibly originally pyrite, which was subsequently replaced by goethite. Abundant fresh and fine pyrrhotite crystals (less than 2 μm) are identified in HA10 and HA12, while samples of HA09 and HA11 contain small amounts of pyrrhotite. The high coercivity of the limestone samples is probably the result of the fine grain size of the pyrrhotite. Because of their growth within calcite crystals, pyrrhotite grains escaped from oxidation, which could be expected as a result of meteoric water circulation.

7. Discussion

7.1. Timing of acquisition for secondary remanent magnetization

Remagnetization of the Permo–Carboniferous sedimentary rocks is confirmed through a negative fold test at 99% confidence level. In situ mean paleomagnetic directions of Silurian and Ordovician rocks are almost identical to those of the Permo–Carboniferous rocks (Fig. 3), indicating that all rocks in the Hayachine–Miyamori district were probably subjected to remagnetization at the same geological time. Because of identical paleomagnetic directions with normal polarity, all 18 sites of the Paleozoic era are grouped into a single mean in situ direction ($D = 321.2°$, $I = 56.5°$, $\alpha_{95} = 5.2°$, $N = 18$).

As shown in Fig. 3b, this secondary westerly direction of the Paleozoic rocks from the Hayachine–Miyamori district is almost parallel to the previously reported secondary directions from the 110–100 Ma Miyako sediments and 129–107 Ma granite of the Kitakami massif [6,7]. This type of similarity strongly indicates that the Paleozoic and Mesozoic rocks of the Kitakami massif were subjected to the same remagnetization event.

The timing of remagnetization in the Kitakami massif can best be estimated by comparing its direction with those of the primary remanent magnetization from northeast Japan (Fig. 3). A list of three paleomagnetic directions has been established from northeast Japan [23,24]: (1) shallow inclination with large counter-clockwise de-

flection in declination from 114–119 Ma Harachiyama and 62–71 Ma Heizaki volcanics; (2) moderately steep inclination with northwesterly declination from welded tuffs of 33–16 Ma; and (3) northerly direction from welded tuffs of 14–5 Ma. These three distinctive directions reflect the tectonic movement of northeast Japan since Mesozoic time, indicating a north–south translation and counter-clockwise tectonic rotation. The mean direction of the secondary magnetization presented here is in good agreement with the paleomagnetic direction of 33–16 Ma ($D = 310.3°$, $I = 57.2°$, $\alpha_{95} = 5.7°$, $N = 32$). We therefore conclude that the remagnetization event probably occurred at a period between 62 and 16 Ma. This important geological event took place prior to the counter-clockwise rotation of northeast Japan as a result of the opening of the Japan Sea.

7.2. Origin of the secondary magnetizations

Rock magnetic studies indicate that the secondary remanent magnetization is carried by magnetite and/or pyrrhotite. The presence of these minerals is also ascertained by microscopic observation.

Magnetite is the carrier of the ChRM in the Ordovician serpentinized ultramafic rocks. Abundant magnetite grains are observed to concentrate in serpentine mesh rims. They are a by-product of serpentinization, which probably occurred in the presence of water at low temperatures (300–500°C). Because olivine crystals remained in the partially serpentinized ultramafic rocks of the Hayachine–Miyamori complexes, serpentinization could have continued. Acquisition of crystallization remanent magnetization (CRM) by the growth of newly formed magnetite (in the course of serpentinization) is the most plausible explanation for the ChRM in the Hayachine–Miyamori ultramafic rocks (e.g. [25]).

The limestone samples carry a remanence associated with pyrrhotite. This is the first time that pyrrhotite has been documented as the main remanence carrier in Paleozoic rocks of the Kitakami massif. Here pyrrhotite was probably formed as a result of an anoxic event, which caused dissolution of the host magnetite [26]. The ChRM in the limestones can therefore be ascribed to CRM acquired as a result of alteration at low temperature instead of any thermochemical activities.

CRM acquired during low-temperature alteration is also reported from the remagnetized Cretaceous granitic rocks in the Kitakami massif [7]. The ChRM of these granitic rocks is determined to be carried by pure-magnetite inclusions in the actinolite part of hornblende. Alteration of hornblende to actinolite occurs at fairly low temperatures.

These facts strongly indicate that the secondary magnetization in the Paleozoic and Mesozoic rocks of the Kitakami massif was acquired as a CRM in a low-temperature condition. This idea is supported by comparison of K–Ar ages of the Hayachine–Miyamori ultramafic rocks (421–484 Ma) and granitic rocks (109–135 Ma) with those inferred from paleontological evidence [2]. No argon loss due to an increase in temperature occurred during the remagnetization event.

The available rock and paleomagnetic parameters from the Haychine–Miyamori serpentinized ultramafic rocks strongly suggest that fluid migration at low temperature is one of the most likely factors in the production of magnetite and the eventual acquisition of secondary magnetization. This phenomenon can be seen in the formula of Komor et al. [27]: olivine+H_2O = serpentine+brucite+magnetite. IThe important role of a tectonically fluid migration is also suggested for diagenetic growth of pyrrhotite [28].

7.3. Tectonic implications

In contrast to the western part, the eastern margin of the Kitakami massif was not imparted by the Cenozoic remagnetization event (Fig. 7). Although remagnetized granitic rocks are widely distributed in the Kitakami massif, granitic rocks with primary NRM were also found in the northeastern part of the massif [29,30]. Harachiyama and Heizaki volcanics (114–119 Ma and 62–71 Ma, respectively), which preserve primary-component magnetization [24], are located in the eastern margin of the massif. The distribution of unremagnetized and remagnetized rocks in the massif implies that the circulation of low-temperature

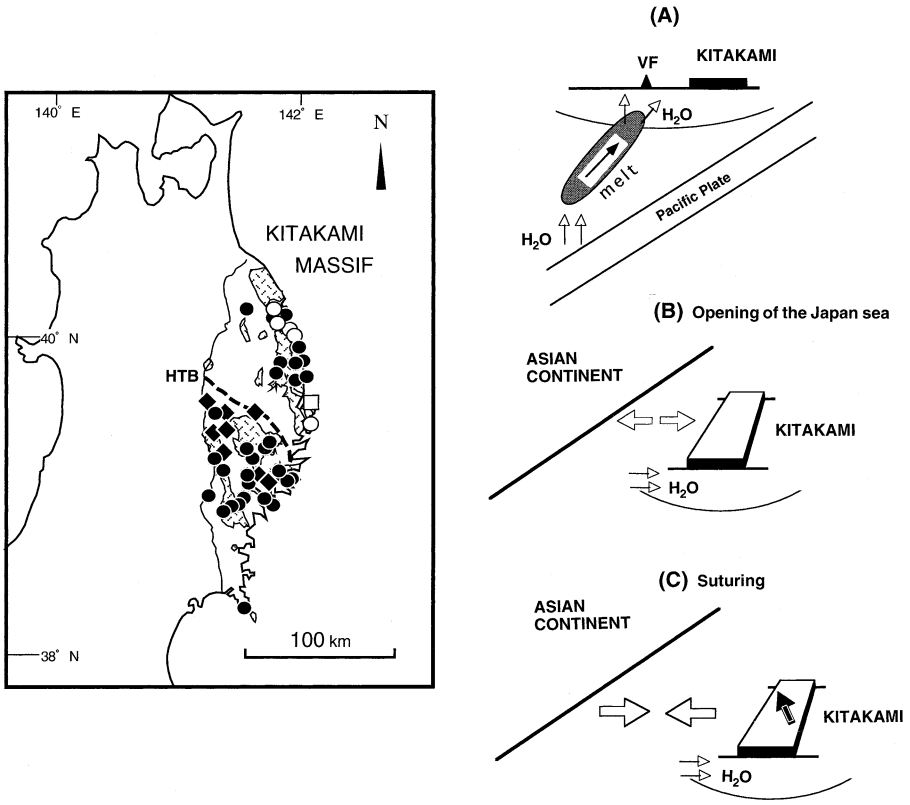


Fig. 7. Distribution of unremagnetized and remagnetized rocks together with models for remagnetization due to water migration. Paleozoic to Cenozoic rocks for the Kitakami massif are classified into two groups: rocks with primary remanent magnetization (open) and remagnetized rocks (closed). Diamond, Paleozoic rocks; circle, Mesozoic rocks; square, Paleogene rocks. HTB, Hayachine tectonic belt. Three models are proposed for water migration. (A) Upward migration of water beneath the western part of the Kitakami massif. VF, Volcanic front. (B) Eastward lateral migration of water toward the western margin of the Kitakami massif either due to opening of the Japan Sea, or (C) due to suturing of the Kitakami massif with the Asian continent after its northward translation (closed arrow).

fluids, which instigated the process of remagnetization, was more vigorous in the western part of the Kitakami massif.

Although numerous mechanisms can be invoked to explain the migration of fluids, they are grouped into three categories from the viewpoints of plate boundaries (Fig. 7): (A) Upward migration of water with melt beneath the western part of the Kitakami massif associated with subduction of the oceanic plate; (B) eastward lateral migration of water toward the western margin of the Kitakami massif during either opening of the Japan Sea; or (C) suturing the Kitakami massif with the Asian continent.

Recent seismological study of northeast Japan reveals that water has been released from the melt beneath the volcanic front after its rise to the lower crust through the mantle wedge [31,32] (Fig. 7A). A significant amount of H_2O-rich fluid could be expelled from the descending slab due to dehydration and eventually absorbed in the melt during its generation as a result of reaction with the convecting material in the mantle wedge (e.g. [33–35]). The presence of a fluid reservoir beneath the volcanic front of northeast Japan has also been suggested by magnetotelluric soundings [36]. Since the volcanic front was once located at the western margin of the Kitakami massif in

Early Miocene (~22 Ma) [37–40], hydrothermal activities may have occurred in association with volcanism. On the other hand, it should be noted that although the Plio–Pleistocene volcanic front is located in the central axis of the northeast Japan arc, no pervasive remagnetization has been observed in this region [23,41,42]. It is therefore argued that upward migration of water can hardly explain the process of remagnetization in the Kitakami massif.

Lateral migration of water associated with the opening of the Japan Sea between the Asian continent and northeast Japan can also be considered to justify the presence of H_2O-rich fluid in the western part of the Kitakami massif (Fig. 7B). Since the opening of the Japan Sea had already started prior to the counter-clockwise rotation of northeast Japan [43], eastward migration of water from the rifting axis of the Japan Sea can be expected between 25 and 15 Ma. Although the opening phenomenon of the Japan Sea also occurred in the area between southwest Japan and the Asian continent, no pervasive remagnetization is observed in the Cretaceous granites and welded tuffs of southwest Japan [30,44,45]. The rifting event accompanied with back arc opening seems to be unfeasible for explanation of the remagnetization of the Kitakami massif.

An alternative interpretation is the Cenozoic suturing of the Kitakami massif with the Asian continent (Fig. 7C), which has recently been postulated by geological and paleomagnetic investigations [9,24,46]. At 71–62 Ma, the Kitakami massif was located at 5° ± 4°N, whereas at the time of remagnetization it was located at 37° ± 5°N. The northward translation of the Kitakami massif and its subsequent suturing with the Asian continent could have provided significant volumes of water to the western margin of the studied massif. Suturing of the Kitakami massif with the Asian continent requires either an oceanic plate subduction or strike-slip fault contact. The fluids expelled from the collision and/or transform margins were probably hosted by serpentines and other rocks in the western part of the Kitakami massif, which resulted in the remagnetization. This phenomenon can be compared with orogen-related remagnetization in the North American platform [1,47,48], remagnetization of the Gangdese batholith along the Indus–Zangbo suture in the Tibetan plateau [49] or strike-slip faulting of the Coast Range ophiolite in California [50].

8. Conclusion

Paleozoic serpentinized ultramafic rocks, mafic rocks, sand–mud stones and limestones are completely remagnetized. The in situ mean direction is $D = 321.2°$, $I = 56.5°$, $\alpha_{95} = 5.2°$, $N = 18$. Remagnetization of these Paleozoic rocks occurred between 62 and 16 Ma. Secondary remanent magnetization of the serpentinized ultramafic rocks is carried by magnetite, while fine-grained pyrrhotite crystals are the carrier of ChRM in limestones. Previous study of the Kitakami granitic rocks reveals that the secondary magnetization is carried by pure-magnetite inclusions in hornblende. These facts imply that remagnetization in the Kitakami massif is generally attributed to a CRM of low-temperature origin. Taking into consideration three different models, it is concluded that the most likely tectonic event responsible for remagnetization in the Kitakami massif is the eastward lateral migration of water. The Cenozoic suturing of the Kitakami massif with the Asian continent is postulated as a plausible tectonic event for fluid migration.

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